

The Genie in the Jmol: Your Wish is My Command!

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Latin American Crystallographic Association
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Acknowledgments

This presentation is
dedicated to George
Hardgrove, Professor of
Chemistry at St. Olaf College
1959-



whose fascination with all things crystallographic
sparked my interest in structure determination
and the beauty of crystallographic symmetry

Acknowledgments

Huge thanks to the Jmol User Group for so many great idea and encouragement over the past 18 years, just a few listed here

Showing 250 ▾ results of 32619

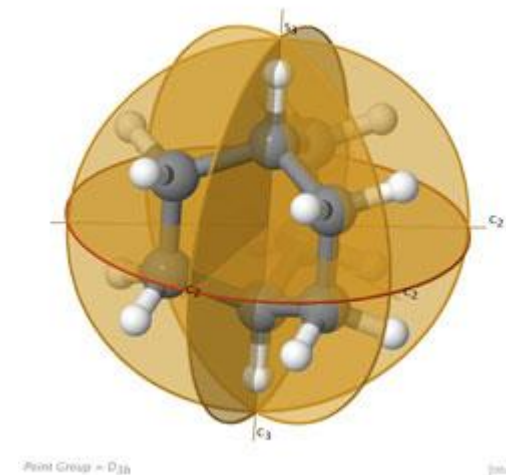
Philip Bays
Norwid Behrnd
I. Camps
Pat Carroll
Martin Foster
Marco Fusè
Victor Rosas Garcia
Nick Greeves
Randall Hall
Ángel Herráez
Michael Howards

Peter Khalifah
John Keller
Martin Kleinschmidt
Matt Kubasik
David Leader
Eric Martz
Nicholas Newell
Mark Perri
Francesco Pietra
Romuald Poteau

Tom Pouce
Jaime Prilusky
Otis Rothenberger
Henry Rzepa
Jörg Saßmannshausen
Joel Sussman
Bruce Tattershall
Kevin Theis
Simon Westrip
Geoge Whitwell
Alex Yang

Acknowledgments

Special thanks to **Dean Johnston** who, at a conference in July of 2008, gave me the idea of adding point group capability to Jmol and used it to develop the **world's best ever web site on point group symmetry**,
<https://symotter.org>



Recent Highlights

Created by Dean Johnston in 2024, this **killer webpage** implements many of the features of Jmol we will be discussing today.



<https://spacegroups.symotter.org>

← → ↻ https://spacegroups.symotter.org/index.html

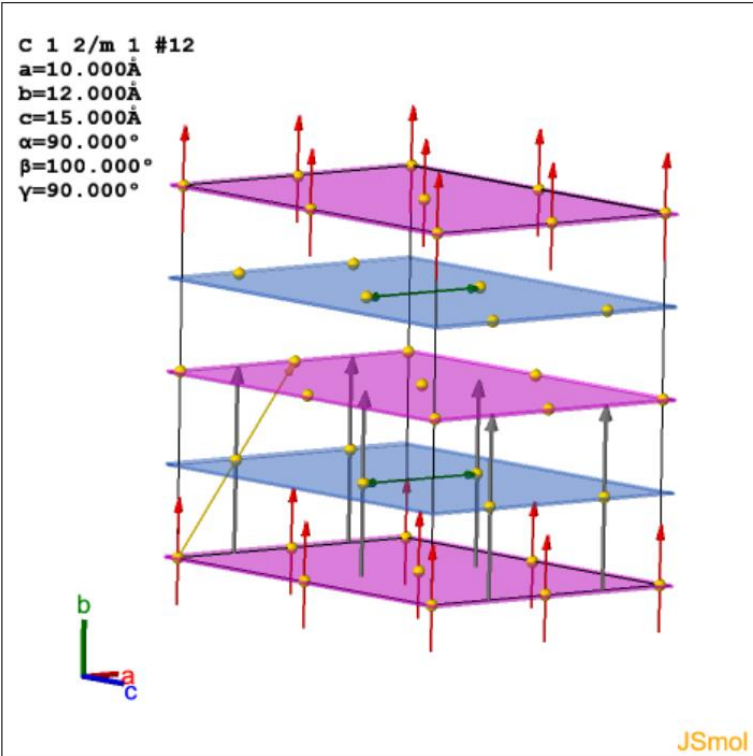
Jmol Space Group Symmetry Visualizer

System: Class:

1: P1
2: PT
3: P2
4: P2₁
5: C2
6: Pm
7: Pc
8: Cm
9: Cc
10: P2/m
11: P2₁/m
12: C2/m
13: P2/c
14: P2₁/c
15: C2/c
16: P222
17: P222₁
18: P2₁2₁2
19: P2₁2₁2₁
20: C222₁
21: C222
22: F222
23: I222

Space Group Visualizer Plane Group Visualizer Subgroup Visualizer Structure Visualizer

C 1 2/m 1 #12
a=10.000Å
b=12.000Å
c=15.000Å
α=90.000°
β=100.000°
γ=90.000°



JSmol

Alignment:

Commands:

Symop Table (C2/m)

SymOps ON SymOps OFF

Inversion Centers

☒ inversion center

Rotation Axes

☒ 2-fold axis
☒ 2-fold screw axis

Glide and Mirror Planes

☒ mirror plane
☒ a-glide plane

Centering Vectors

☒ centering vector

Wyckoff Table

mult	label	desc	geom		
8	j	1	general	add	del
4	i	m	plane	add	del
4	h	2	line	add	del

Recent Highlights

Just published!

Bilbao Crystallographic
Server, Cambridge
Structural Database,
VESTA, and *Jmol*.



<http://doi.org/10.1107/S1600576724007659>



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teaching and education

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<https://doi.org/10.1107/S1600576724007659>

OPEN  ACCESS  

Free tools for crystallographic symmetry handling and visualization

**Gemma de la Flor,^{a*} Mois I. Aroyo,^b Ilaria Gimondi,^c
Suzanna C. Ward,^c  Koichi Momma,^d  Robert M.
Hanson^e  and Leopoldo Suescun^f **



Recent Highlights

Just published!

Bilbao Crystallographic
Server, Cambridge
Structural Database,
VESTA, and *Jmol*.



<http://doi.org/10.1107/S1600576724007659>

Table 1

Jmol load command options relating to crystallographic symmetry

Table 2

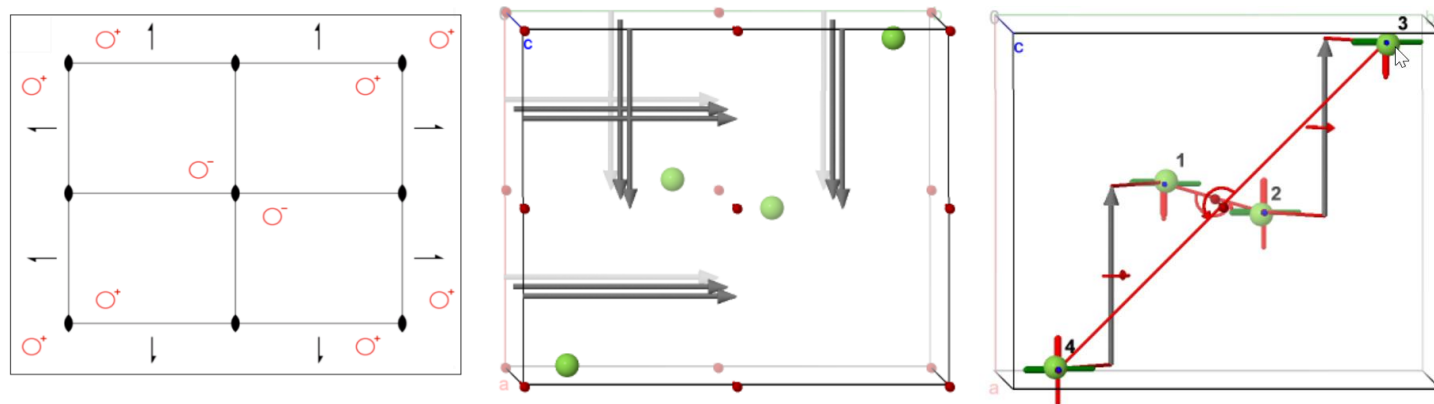
Jmol modelkit commands relating to crystallography

Table 3

Various questions and how to answer them using *Jmol* scripting functions

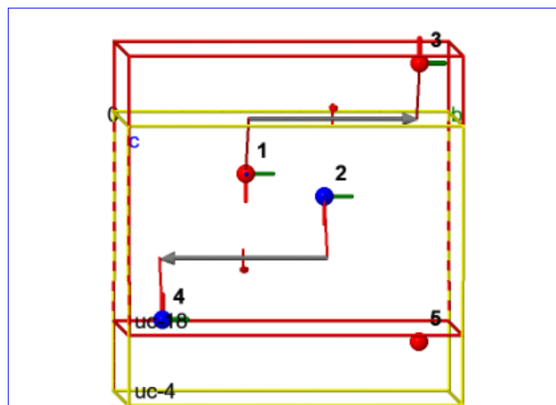
Table 4

Representative websites incorporating *JSmol* in the area of crystallography



Recent Highlights

Includes a supporting website that allows you to experiment with Jmol and to **recreate all the figures in this section of the paper yourself** using JSmol, step by step.



NCI(small molecules) Search

[open...](#) [console](#) [image](#) [display](#) [faster](#) [sharper](#) [large](#) [small](#)
platformSpeed: 8 7 6 5 4 3 2 1
info [show](#) [clear](#) [hide](#)
[spacefill](#) [wire](#) [ball&stick](#) [cartoons](#) [fancy](#) [not flat](#)
[color](#) [atomno](#) [color](#) [cpk](#) [color](#) [structure](#)
[isosurface](#) [vdw](#) [off](#) [mep](#) [translucent](#) [opaque](#)
labels [on](#) [off](#) [echo](#) [larger](#) [smaller](#)

Figure [script "figure16.e2.png"](#)

[zap;modelkit spacegroup 18; rotate x 2; rotate y 2](#) Illustrating the a different transformation pathway from P 21 21 21 (No.18) to P 21 (No. 4).

- [modelkit add "F" {0.460 0.383 0.288/1}](#)
- [draw uc18 unitcell color red "uc-18"](#)
- [modelkit spacegroup "18>a,b,c;1/4,0,0>4"](#)
- [packed](#)
- [translate y -5](#)
- [draw uc4 unitcell color yellow "uc-4"](#)
- [label "%\[atomno\]";color labels black](#)
- [set labelOffset 4 4;font label 16 bold](#)
- [color property site](#)
- [draw s23 symop @1 @3](#)
- [draw s41 symop @2 @4](#)
- [set frank off;set showunitcellinfo false](#)
- [set picking dragatom](#)
- [prompt "BCS data used: " + spacegroup\(18,4,"subgroups"\)](#)

more examples

[zap;modelkit spacegroup 18](#)

Illustrating the differences in symmetry between P 21 21 21 (No. 18) and its subgroup P 21 (No. 4).

Quick Introduction to Jmol

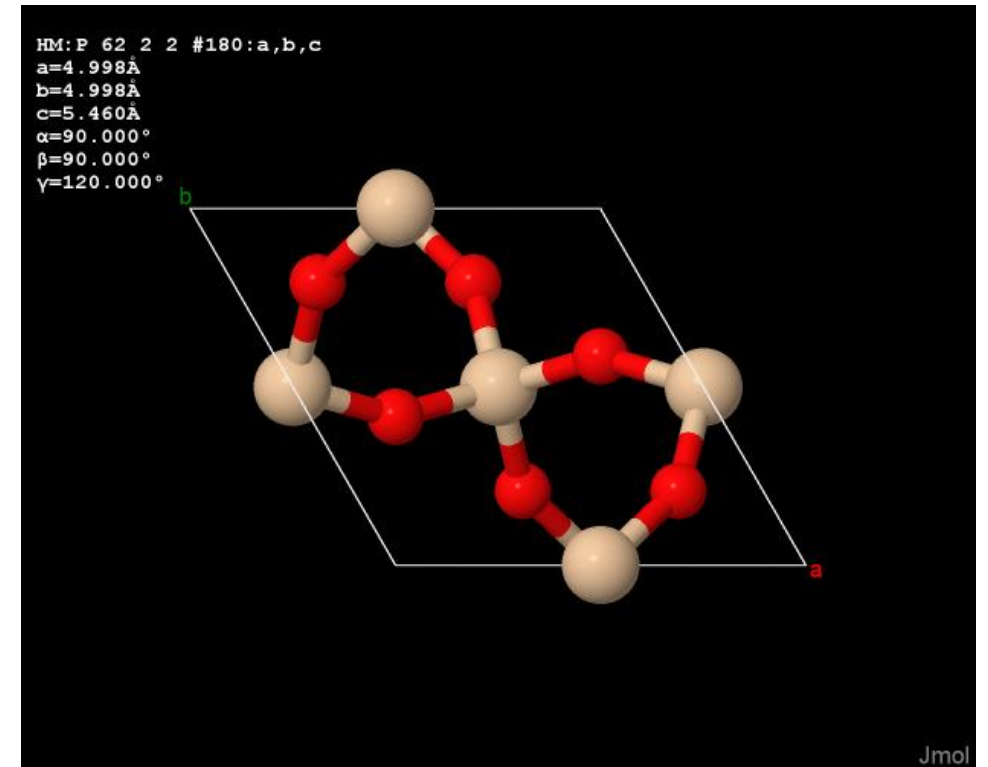
Mission: The high-quality, real-time visualization
and analysis of molecular structure, dynamics,
and energetics.

Jmol is:

- open source Java
- automatically transpiled to JavaScript
- cross-disciplinary
- actively being developed

Jmol is not:

- a commercial enterprise
- a quantum computational package

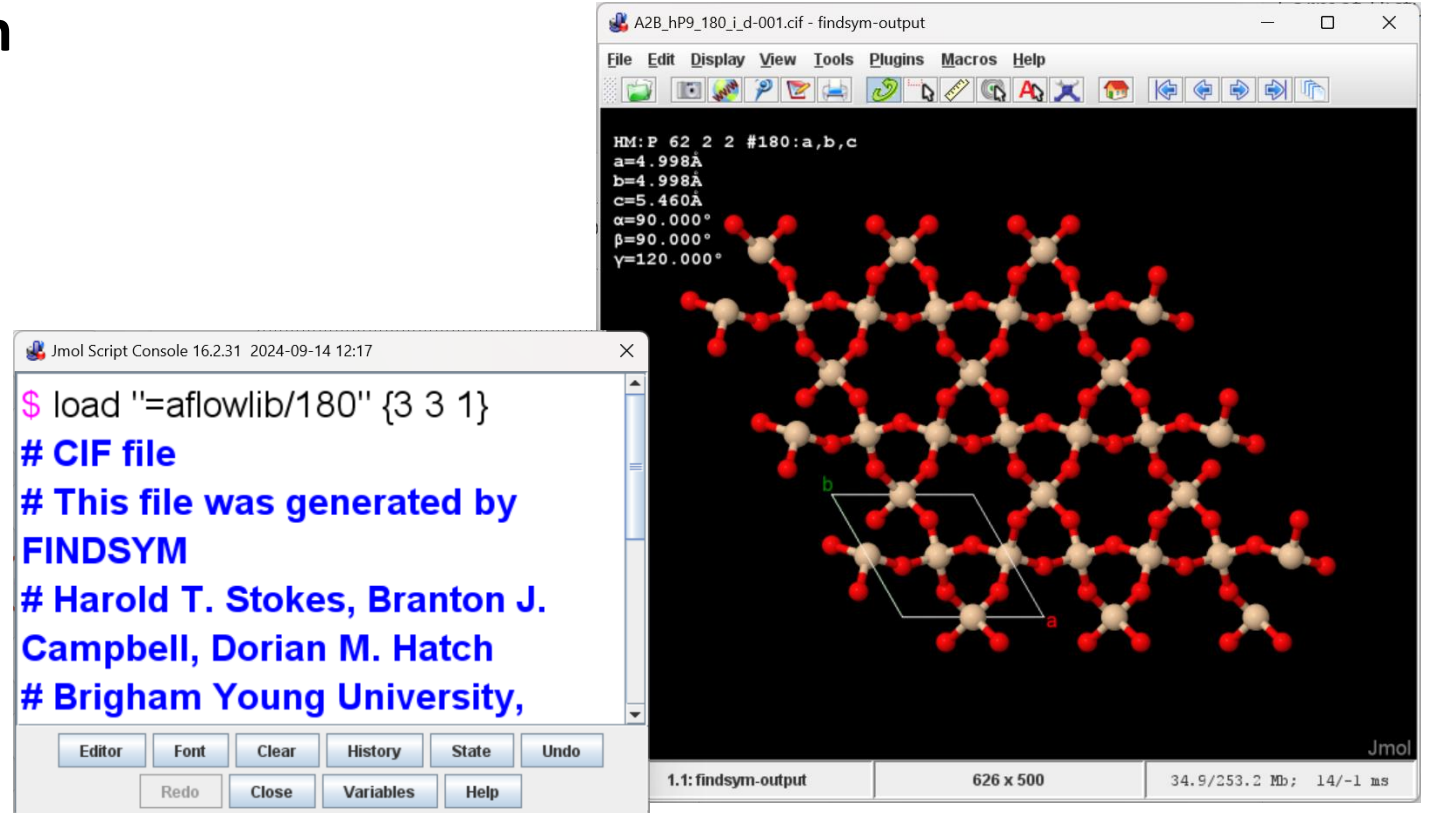


Quick Introduction to Jmol

Configurations:

Jmol.jar Java Application

fast, full service,
Allows drag&drop

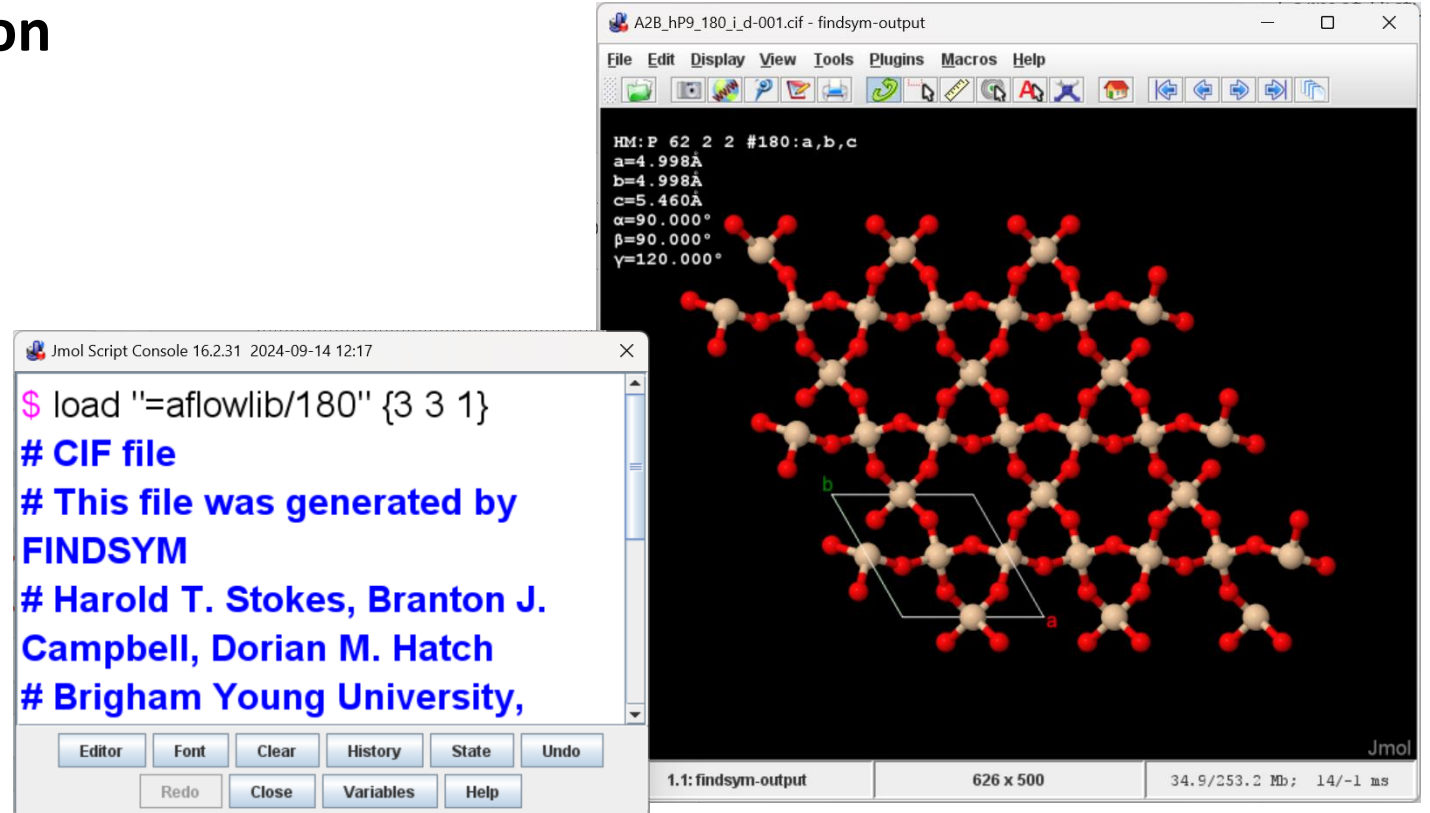


Quick Introduction to Jmol

Configurations:

JmolD.jar Java Application

Full double precision



Quick Introduction to Jmol

Configurations:

JmolData.jar and JmolDataD.jar

command-line interface

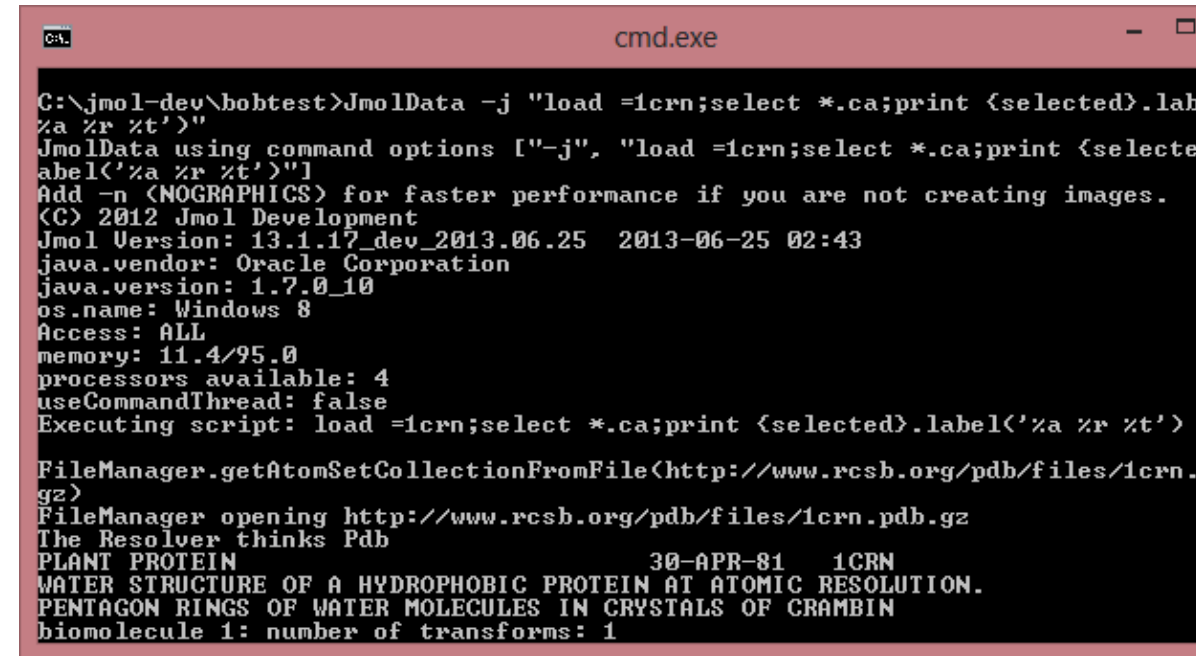
fully scriptable

runs “headless” (no user interface)

can create images and do analyses

allows for automated workflow

can be used as server-side app



```
C:\jmol-dev\bobtest>JmolData -j "load =1crn;select *.ca;print {selected}.label '%a %r %t'"
JmolData using command options ["-j", "load =1crn;select *.ca;print {selected}.label '%a %r %t'"]
Add -n (NOGRAPHICS) for faster performance if you are not creating images.
(C) 2012 Jmol Development
Jmol Version: 13.1.17_dev_2013.06.25 2013-06-25 02:43
java.vendor: Oracle Corporation
java.version: 1.7.0_10
os.name: Windows 8
Access: ALL
memory: 11.4/95.0
processors available: 4
useCommandThread: false
Executing script: load =1crn;select *.ca;print {selected}.label '%a %r %t'

FileManager.getAtomSetCollectionFromFile(http://www.rcsb.org/pdb/files/1crn.pdb.gz)
FileManager opening http://www.rcsb.org/pdb/files/1crn.pdb.gz
The Resolver thinks Pdb
PLANT PROTEIN 30-APR-81 1CRN
WATER STRUCTURE OF A HYDROPHOBIC PROTEIN AT ATOMIC RESOLUTION.
PENTAGON RINGS OF WATER MOLECULES IN CRYSTALS OF CRAMBIN
biomolecule 1: number of transforms: 1
```

Quick Introduction to Jmol

Configurations:

JavaScript App

double precision

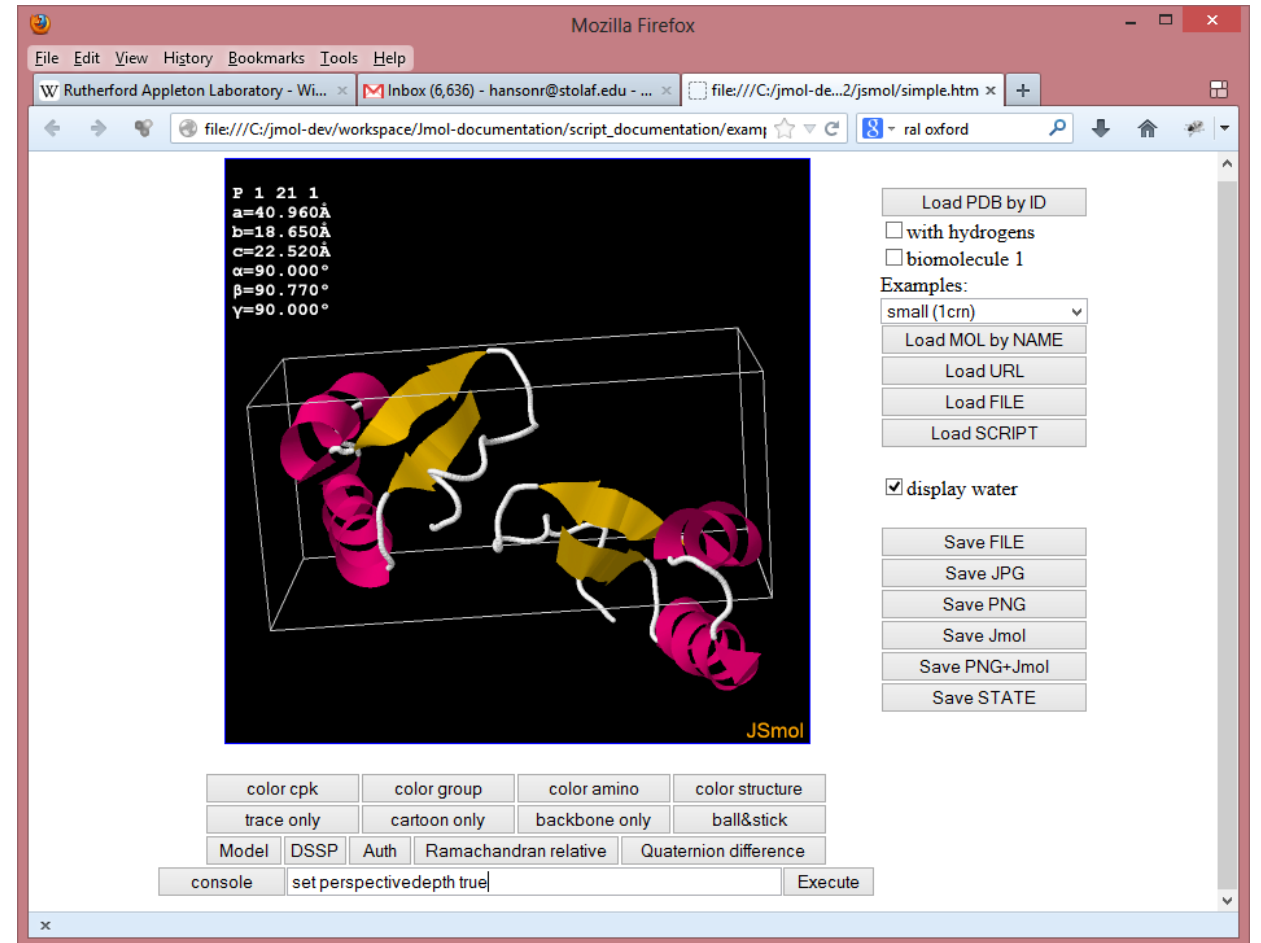
identical to JmolD.jar

works in all popular browsers,
including

Safari for the iPad, android
phones

allows drag&drop

somewhat slower than Java



I wish...

I wish...

Jmol - hansonr@stolaf.edu

Idea:

Name:

Email:

I wish...

...Jmol could quickly load example files from any space group.

I wish...

...Jmol could quickly load example files from any space group.

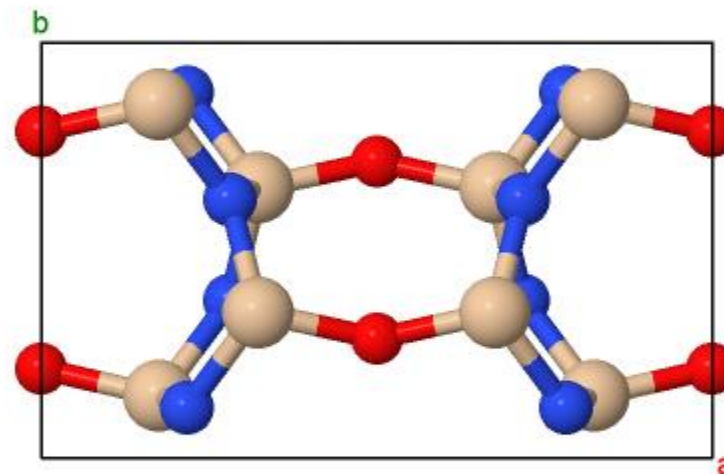
load =afflowlib/36.6 packed

Loads a structure from the AFLOW Encyclopedia of Crystallographic Prototypes

All space groups are represented by at least one structure.

<https://afflowlib.org/prototype-encyclopedia>

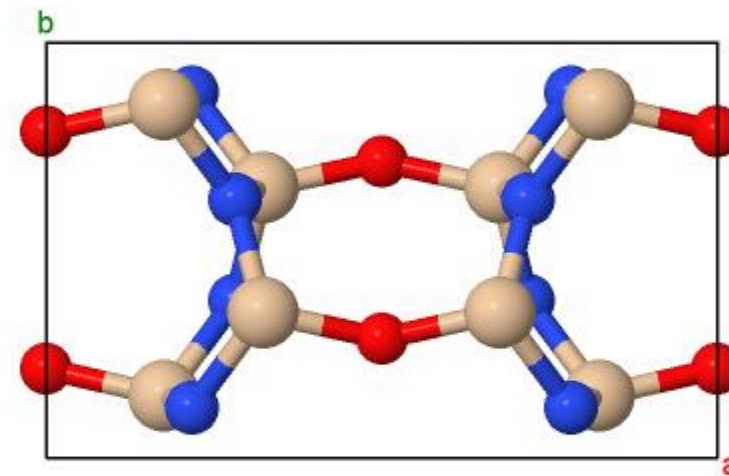
```
HM:C m c 21 #36:a,b,c  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

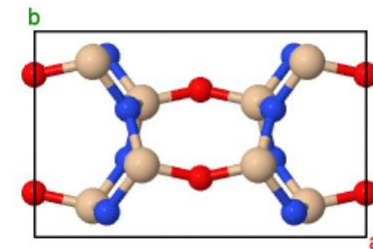
...I could build a crystal from scratch using Jmol.

```
HM:C m c 21 #36:a,b,c  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

...I could build a crystal from scratch using Jmol.



```
load =afflowlib/36.6
```

```
show unitcell
```

```
a=8.843, b=5.473, c=4.835,
```

```
alpha=90.0, beta=90.0, gamma=90.0
```

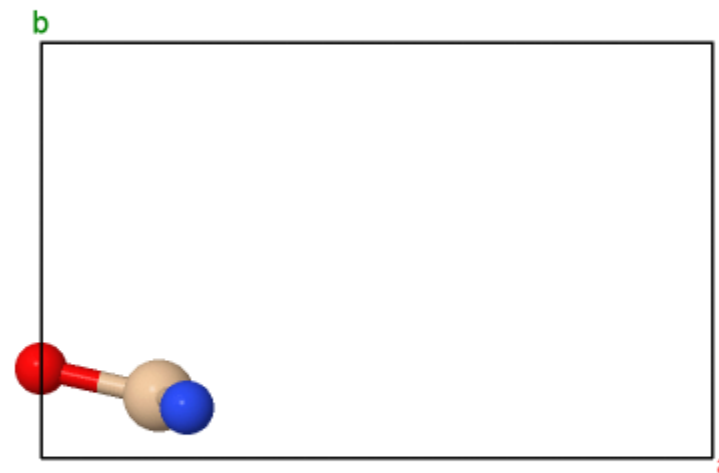
```
print {*}.label("%e %6.3[fxyz]")
```

```
O 0.000 0.214 0.230
```

```
N 0.218 0.121 0.642
```

```
Si 0.176 0.151 0.290
```

```
HM:C m c 21 #36:a,b,c  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



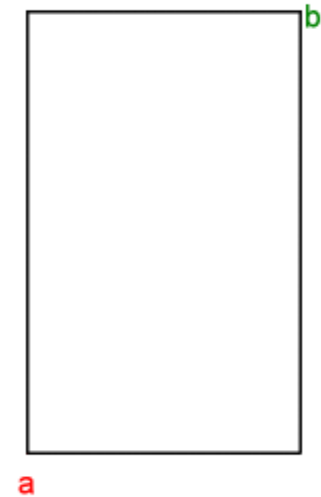
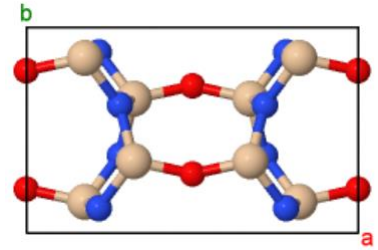
I wish...

...I could build a crystal from scratch using Jmol.

modelkit zap spacegroup 36

unitcell [8.843 5.473 4.835 90 90 90]

```
C m c 21 #36
a=8.843Å
b=5.473Å
c=4.835Å
α=90.000°
β=90.000°
γ=90.000°
```



I wish...

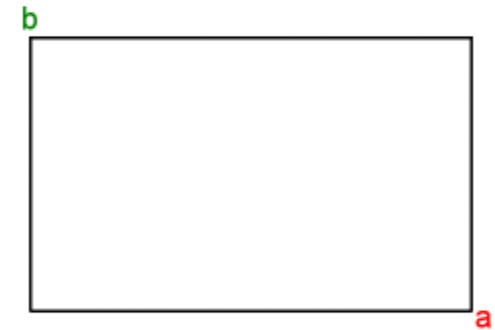
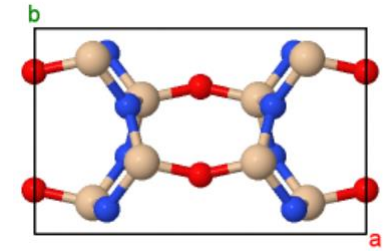
...I could build a crystal from scratch using Jmol.

modelkit zap spacegroup 36

unitcell [8.843 5.473 4.835 90 90 90]

moveto axis c4

```
C m c 21 #36
a=8.843Å
b=5.473Å
c=4.835Å
α=90.000°
β=90.000°
γ=90.000°
```



I wish...

...I could build a crystal from scratch using Jmol.

modelkit zap spacegroup 36

unitcell [8.843 5.473 4.835 90 90 90]

moveto axis c4

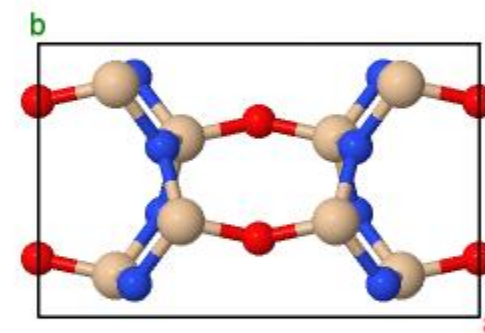
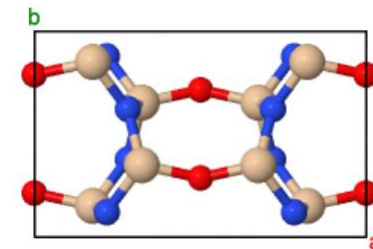
modelkit add O {0.000 0.214 0.230/1} packed

modelkit add N {0.218 0.121 0.642/1} packed

modelkit add Si {0.176 0.151 0.290/1} packed

connect

```
C m c 21 #36
a=8.843Å
b=5.473Å
c=4.835Å
α=90.000°
β=90.000°
γ=90.000°
```



I wish...

...I could move atoms around in a crystal structure.

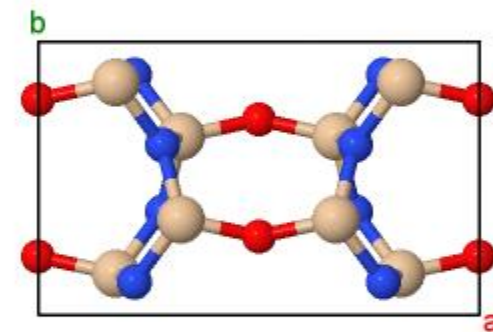
I wish...

...I could move atoms around in a crystal structure.

load =afflowlib/36.6 packed

set picking dragatom

```
C m c 21 #36  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



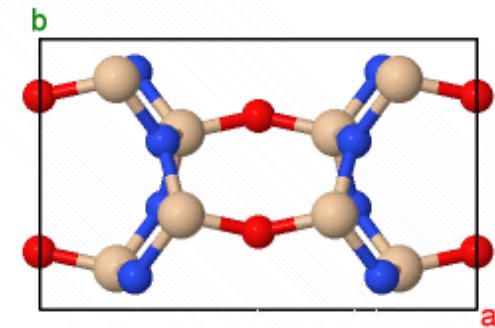
I wish...

...I could move atoms around in a crystal structure.

load =afflowlib/36.6 packed

set picking dragatom

```
C m c 21 #36
a=8.843Å
b=5.473Å
c=4.835Å
α=90.000°
β=90.000°
γ=90.000°
```



I wish...

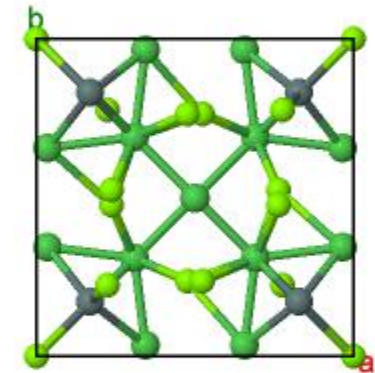
...Jmol could switch among space group settings for a loaded model.

I wish...

...Jmol could switch among space group settings for a loaded model.

load =aflowlib/101 packed;zoom 60

```
HM:P 42 c m #101:a,b,c  
a=9.851Å  
b=9.851Å  
c=6.868Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

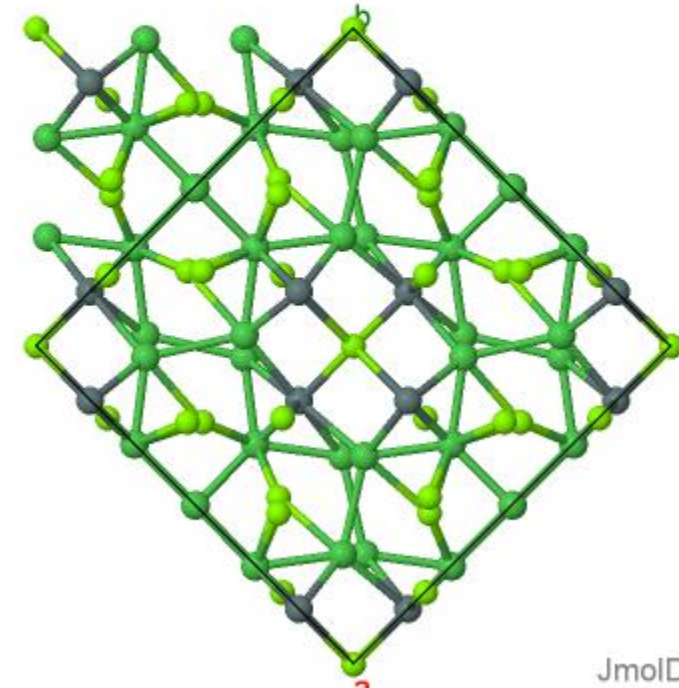
...Jmol could switch among space group settings for a loaded model.

load =aflowlib/101 packed;zoom 60

modelkit spacegroup "C 42 m c" packed

connect

```
C 42 m c #101:a-b,a+b,c  
a=13.931Å  
b=13.931Å  
c=6.868Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

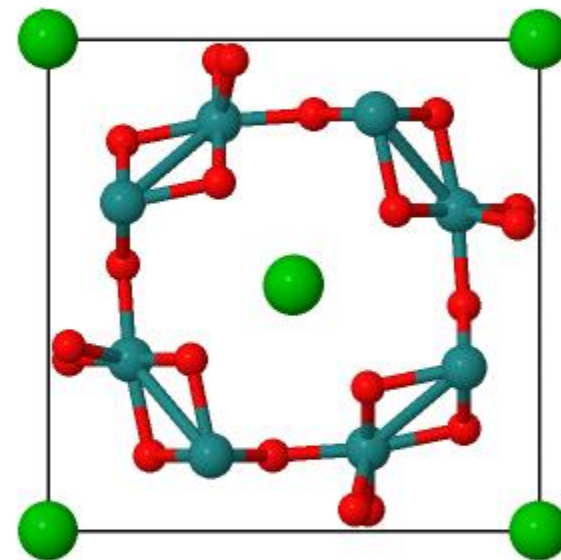
...Jmol could identify a model's space group based on its symmetry.

I wish...

...Jmol could identify a model's space group based on its symmetry.

load =aflowlib/75 packed
modelkit spacegroup p1

```
P 1 #1  
a=9.888Å  
b=9.888Å  
c=9.121Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



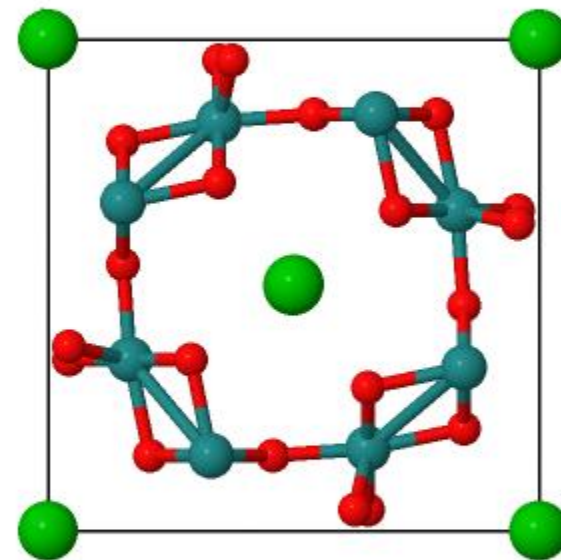
I wish...

...Jmol could identify a model's space group based on its symmetry.

```
load =afwlib/75 packed  
modelkit spcaegroup p1  
modelkit spacegroup
```

75 HM:P 4 #75:a,b,c

```
P 4 #75  
a=9.888Å  
b=9.888Å  
c=9.121Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

...I could show a space group's symmetry elements in Jmol.

I wish...

...I could show a space group's symmetry elements in Jmol.

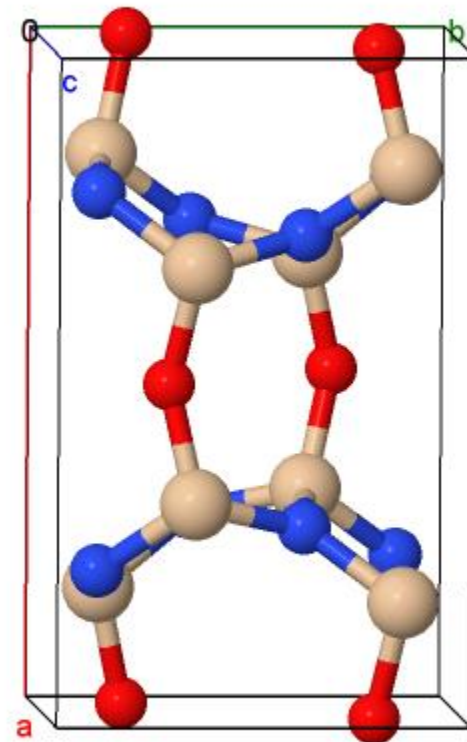
load =afLOWlib/36.6 packed

moveto axis c1

rotate x 5

rotate y 5

```
HM:C m c 21 #36:a,b,c  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

...I could show a space group's symmetry elements in Jmol.

load =afwlib/36.6 packed

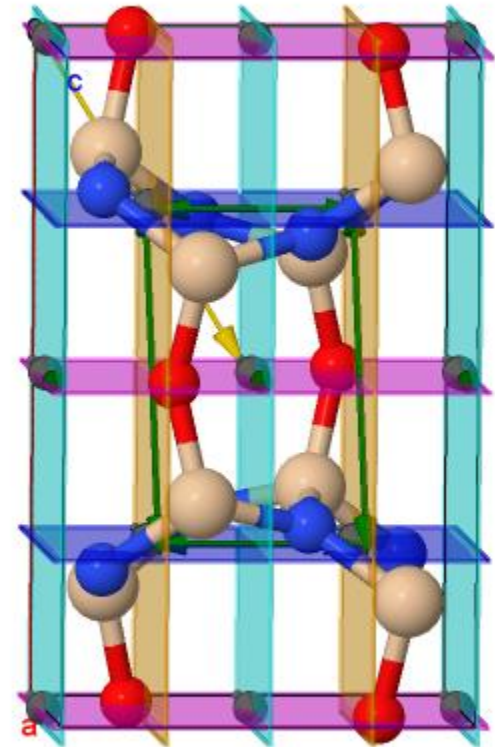
moveto axis c1

rotate x 5

rotate y 5

draw spacegroup all

```
HM:C m c 21 #36:a,b,c  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

...I could show just ***some*** of a space group's symmetry elements in Jmol.

I wish...

...I could show just *some* of a space group's symmetry elements in Jmol.

load =afflowlib/36.6 packed

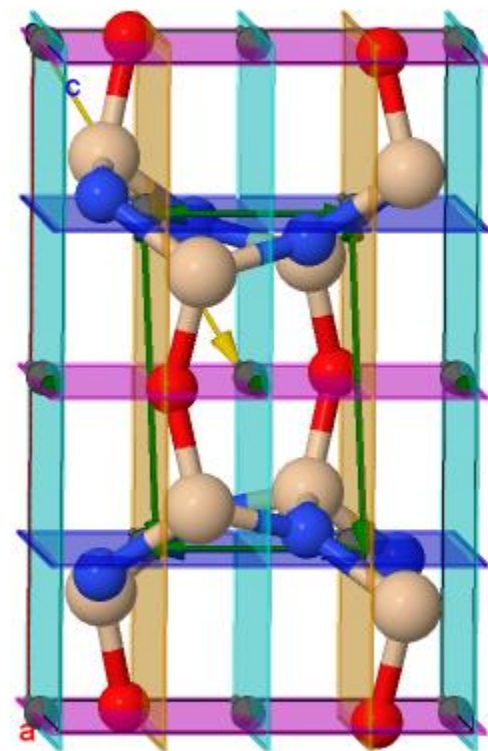
moveto axis c1

rotate x 5

rotate y 5

draw spacegroup all

```
HM:C m c 21 #36:a,b,c  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



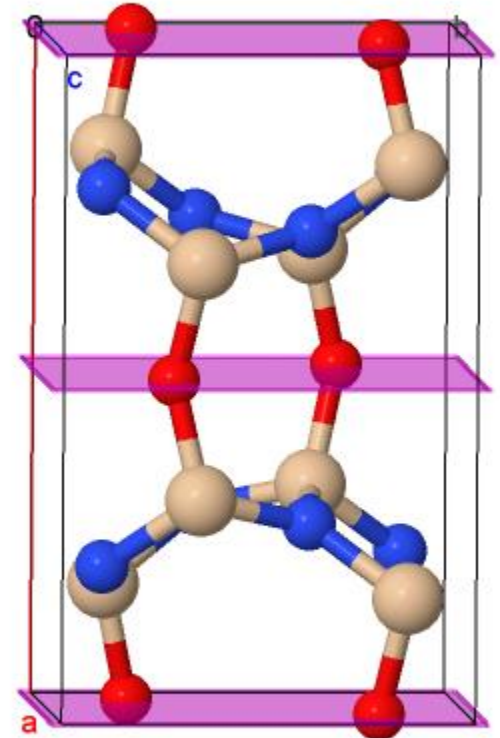
I wish...

...I could show just ***some*** of a space group's symmetry elements in Jmol.

draw off

draw ****mirror**** on

```
HM:C m c 21 #36:a,b,c  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

...I could show just ***some*** of a space group's symmetry elements in Jmol.

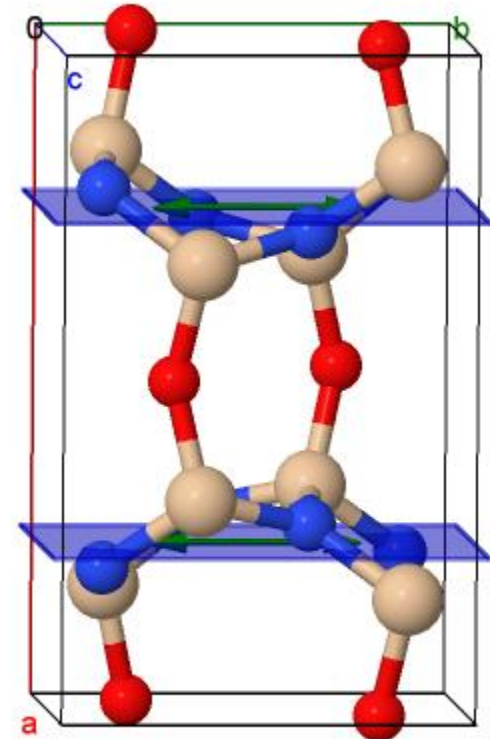
draw off

draw *mirror* on

draw off

draw ***b-g*** on

```
HM:C m c 21 #36:a,b,c  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

...I could show just ***some*** of a space group's symmetry elements in Jmol.

draw off

draw *mirror* on

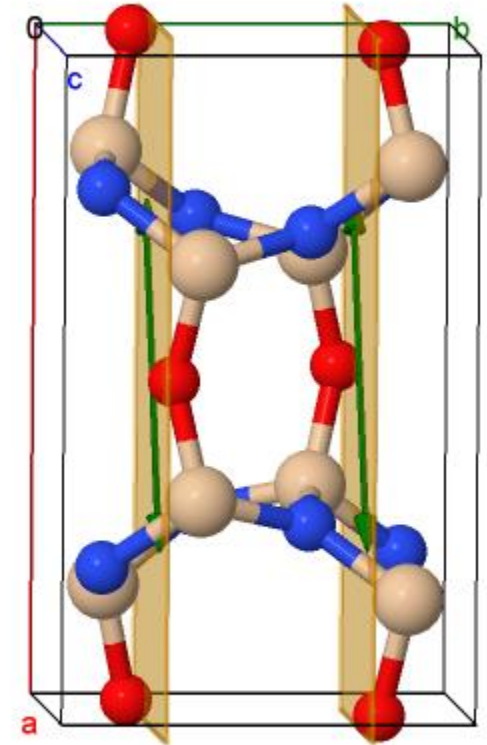
draw off

draw *b-g* on

draw off

draw *n-g* on

HM:C m c 21 #36:a,b,c
a=8.843Å
b=5.473Å
c=4.835Å
 $\alpha=90.000^\circ$
 $\beta=90.000^\circ$
 $\gamma=90.000^\circ$



I wish...

...I could show just ***some*** of a space group's symmetry elements in Jmol.

draw off

draw *mirror* on

draw off

draw *b-g* on

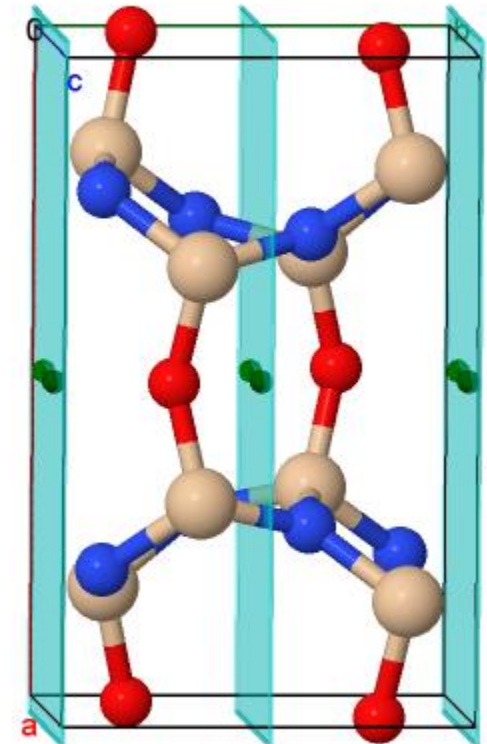
draw off

draw *n-g* on

draw off

draw *c-g* on

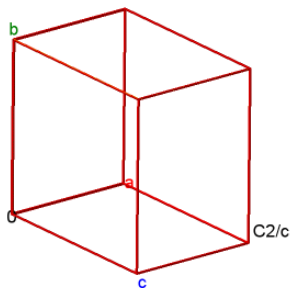
```
HM:C m c 21 #36:a,b,c  
a=8.843Å  
b=5.473Å  
c=4.835Å  
α=90.000°  
β=90.000°  
γ=90.000°
```



I wish...

...Jmol could...

C 1 2/c 1 #15
a=10.000Å
b=12.000Å
c=15.000Å
α=90.000°
β=100.000°
γ=90.000°



JmolD

?

Your turn!

